

27	C	Option A : Incorrect. If the frequency of light is sufficiently high, even at extremely low intensities there will be emission of photoelectrons. Option B : Incorrect. From Einstein's photoelectric effect equation, the relationship between frequency and stopping potential is not directly proportional.

		Option D: Increasing the intensity of the incident photons will not change the energy of each photon, it just increases the no. of photons per unit time incident on the metal. Hence, the energy of the photoelectrons emitted will not change as the interaction between each photon and electron remains the same.
28	B	$K_{\text{max}} = 0$ eV occurs when $\lambda = 300$ nm. Therefore, the threshold wavelength is 300 nm.